

User Study

Instructions for Grading. **PLEASE READ CAREFULLY.**

Overview

Imagine you are a user interacting with a chatbot service (e.g., Gemini, ChatGPT, or DeepSeek) for help with programming tasks. On each page, you will see:

- **A Question:** This could be a vague user query or a code snippet requiring clarification or completion. Questions containing code will be displayed in code blocks for clarity.
- **Two Responses:** labeled Answer A and Answer B

Your Task

Please evaluate Answer A and Answer B based on the following criteria:

- **Precision and Focus:** How directly the response targets the question's core issue, using clear guidance, examples, or code.
- **Contextual Fit:** How well the response matches the user's programming goal (e.g., debugging, feature implementation).
- **Answer Faithfulness:** How effectively the response provides actionable steps or code to achieve the user's desired outcome.
- **Correctness:** Correctness in LLM responses refers to the accuracy and factual validity of the information, code, or guidance provided, ensuring it is free from errors and aligns with the user's query and intended outcome.

How to Rate

For each question, you will provide two types of ratings:

1. **Preference Rating (1–5 scale):**
 - **"1"**: Answer A is better, based on the criteria above.
 - **"5"**: Answer B is better, based on the criteria above.
 - **"3"**: Both answers are equally good, based on the criteria above.

Note:

- **Incomplete Responses:** Some responses may appear cut off due to a 256-word limit imposed on the chatbots. Please grade only the content provided, ignoring any abrupt endings or non-English text (e.g., stray characters).
- **Code Formatting:** Questions and responses containing code will be presented in code blocks or artifact tags for clarity. Ensure you review these carefully for correctness and relevance.
- **Consistency:** Try to apply the same standards across all evaluations to ensure fair comparisons.

Question 1

I'm having trouble with my project.

Answer A:

Okay, I understand you're having trouble with your project. That's really tough, and I'm here to help! To give you the best advice, I need a little more information. Let's break this down. First, let's try to understand the situation. Could you tell me a bit more about what's going on? Let's start with some general questions, and then we can get more specific:

1. What kind of project is it? (e.g., school assignment, personal project, work project, creative endeavor?)
2. What specifically is the problem? (Be as detailed as possible. For example: "I'm stuck on the introduction," or "I don't know where to start," or "I'm getting a specific error message.")
3. What have you already tried to solve the problem? (This helps me avoid suggesting things you've already done.)
4. What's the deadline? (Knowing the deadline helps us prioritize.)... 5. Can you give me a little more context? (e.g., "I've been working on this for weeks," or "I

Answer B:

“Extra context:

User Question: I'm having trouble with my project.

AI clarification question: Okay, let's figure out what's causing the trouble. To help me understand the problem better, could you tell me: What specific part of the project are you struggling with? (e.g., coding, design, organization,)

User follow up interaction: I'm working on a Python web app, and the user authentication part is failing with a 500 error.”

AI Final answer:

Okay, that's a really common problem! A 500 error with a user authentication system in a Python web app is a significant hurdle. Let's break this down.

To help me understand the issue better, let's start with a few key questions:

1. What's the exact error message you're seeing? (Copy and paste the *entire* error message, including any traceback information. This is incredibly helpful.)
2. What is the user authentication system *supposed* to do? (e.g., register a new user, log in, reset a password?)
3. What have you tried so far to troubleshoot it? (Have you checked the logs? Debugged the code? Modified any configurations?)
4. Can you give me a brief overview of the code you're currently using for the authentication part? (Even a high-level description of what the code does and where it's located will be useful.)

Let's start with the error message, and then we'll build from there. Don't worry if you don't have all the answers right away – just give me as much information as you can!

How to rate:

"1": strongly prefer Answer A

"5": strongly prefer Answer B

"3": both answers are equally good

Precision and Focus (1-5): 3

Contextual Fit (1-5): 3

Answer Faithfulness (1-5): 3

Correctness (1-5): 3

Question 2

```
Python
import numpy as np
def compute_matrix():
    return np.array([, [3,4]])
compute_matrix()
```

Answer A:

```
Python
import numpy as np
def compute_matrix():
    return np.array([[1,,]])
compute_matrix(compute_matrix)
```

Answer B:

“Extra context:

User Question:

```
Python
import numpy as np
def compute_matrix():
    return np.array([, [3,4]])
compute_matrix()
```

AI clarification question: What kind of matrix are you hoping to compute? For example, are you looking for the determinant, inverse, or something else?

User follow up interaction: I want to compute the determinant of the 2x2 matrix.”

AI Final answer:

Okay, great! You want to compute the determinant of a 2x2 matrix.
Let's compute the determinant of the matrix:

Python

```
import numpy as np
```

```
def compute_matrix():  
    return np.linalg.det(np.array([[1, 2], [3, 4]]))
```

The determinant of the matrix $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ is:

$$\det(A) = (1 * 4) - (2 * 3) = 4 - 6 = -2$$

Therefore, the determinant of the matrix is -2.

Do you want to try another matrix, or would you like to explore other matrix operations (like the inverse)?

How to rate:

"1": strongly prefer Answer A

"5": strongly prefer Answer B

"3": both answers are equally good

Precision and Focus (1-5): 5

Contextual Fit (1-5): 5

Answer Faithfulness (1-5): 5

Correctness (1-5): 5

Question 3

Python

```
def connect_to_db():
    connection = None
    # TODO: Establish database connection
    return connection
```

Answer A:

Python

```
def connect_to_db():
    """... Establishes a connection to a database (simulated in this
    example).

    Returns:
        Connection object: A database connection object if successful,
        None otherwise.
    """
    try:
        connection = "Database Connection String" # Replace with your actual
        connection string
        print("Database connection established.")
        return connection
    except Exception as e:
        print(f"Error establishing database connection: {e}")
        return None
```

Explanation and Important Considerations:

1. Simulated Database Connection: The code currently uses a placeholder string "Database Connection String". You *must* replace this with the actual connection string that you'll use to connect to your database (e.g., a MySQL connection string, a PostgreSQL connection string, or a SQLite database connection string). The format will vary depending on the database you're using.
2. try...except Block: This is *crucial* for robust error handling. Database connections can fail for various reasons (network issues, incorrect credentials, database server unavailable, etc.). The \try

Answer B:

“Extra context:

User Question:

```
Python
def connect_to_db():
    connection = None
    # TODO: Establish database connection
    return connection
```

AI clarification question: *What type of database are you trying to connect to, and what specific error are you expecting to encounter when you try to connect? For example, are you getting a connection error, or is it just a general feeling that the connection isn't working?*

User follow up interaction: *I'm trying to connect to a PostgreSQL database and getting a 'DatabaseError: connection timeout' error.”*

AI Final answer:

Okay, that's a very common issue when connecting to PostgreSQL. A "DatabaseError: connection timeout" means your code is attempting to connect to the database, but the connection isn't being established quickly enough. Here's a breakdown of what's happening and how to troubleshoot it, along with potential causes and solutions:

Understanding the Problem

The connection timeout error indicates that the database server isn't responding to your connection request within a reasonable timeframe. It's not a general "connection issue" – it's a timing problem. The database server might be overloaded, the network might be slow, or there could be a firewall blocking the connection.

Troubleshooting Steps

1. Check Your Network Connectivity:
 - Ping: From your machine, ping <your_database_host> (e.g., ping mydatabase.example.com). If the ping fails, you have a basic network connectivity problem.
 - Traceroute: traceroute <your_database_host> This will show you the path your packets are taking to the database server. Look for any hops where the latency is very high. This can pinpoint where the problem lies.
2. Verify Database Server Status: Make sure the database server is running and accepting connections.
3. Check Firewall Settings: Ensure that firewalls on your machine, network, or server aren't blocking the connection port.
4. Review Connection String: Confirm that your connection string is correct and uses the right host, port, username, and password.

5. Increase Timeout Settings: Sometimes increasing the timeout value in your database client can help.

If you want, I can help you write code to implement these checks or help debug your connection string.

How to rate:

"1": strongly prefer Answer A

"5": strongly prefer Answer B

"3": both answers are equally good

Precision and Focus (1-5): 5

Contextual Fit (1-5): 5

Answer Faithfulness (1-5): 4

Correctness (1-5): 4

Question 4

```
Python
def calculate_sum(a, b):
    return a + b
result = calculate_sum(5, 1)
```

Answer A:

```
Python
def calculate_sum(a, b):
    return a + b

result = calculate_sum(5, )
print(result)
```

Explanation:

1. `def calculate_sum(a, b):`: This defines a function named `calculate_sum` that takes two arguments, `a` and `b`.
2. `return a + b`: Inside the function, this line calculates the sum of `a` and `b` and returns the result.
3. `result = calculate_sum(5, )`: This line calls the `calculate_sum` function with `a = 5` and `b =` (an empty string). The function will return `5 +` (which is 5). This returned value is then assigned to the variable `result`.
4. `print(result)`: This line prints the value of the `result` variable, which is 5.

Output:

5

Answer B:

“Extra context:

User Question:

```
Python
def calculate_sum(a, b):
    return a + b
result = calculate_sum(5, 1)
```

AI clarification question: *What is the expected behavior or purpose of the function calculate_sum? Specifically, what value should the function return if it's called with the arguments a=5 and b=1?*

User follow up interaction: *I expect it to return the sum of two numbers a and b."*

AI Final Answer:

You are absolutely correct!

The function calculate_sum(a, b) is designed to take two numbers, a and b, as input and return their sum.

If a=5 and b=1, the function will return $5 + 1 = 6$.

Your explanation is perfect – it accurately describes the function's purpose and the expected behavior. Well done!

How to rate:

"1": strongly prefer Answer A

"5": strongly prefer Answer B

"3": both answers are equally good

Precision and Focus (1-5): 5

Contextual Fit (1-5): 5

Answer Faithfulness (1-5): 5

Correctness (1-5): 5

Question 5

I don't understand why it's not functioning.

Answer A:

Okay, let's troubleshoot this! To help me figure out why your device isn't working, I need more information. "It's not functioning" is a very broad statement. Let's break it down. Please tell me about:

1. What are you trying to do? (Be specific!)

- What device are you having trouble with? (e.g., phone, computer, printer, smart home device, etc.)
- What were you trying to do *before* it stopped working? (e.g., send a text, print a document, play music, etc.)

2. What exactly is happening (or *not* happening)?

... * Is it completely dead? (No power at all?)

- Is it slow? (Does it take a long time to respond?)
- Is it unresponsive? (Does it freeze or just not do anything?)
- Are there specific errors? (If so, what do they say? Write them down exactly.) For example: "Error 404," "Cannot connect to Wi-Fi," "The screen is blank," etc.)

Answer B:

“Extra context:

User Question: *I don't understand why it's not functioning.*

AI clarification question: *Please provide the prompt you're struggling with. I need to see the prompt itself to understand why it's not working. 😊*

User follow up interaction: *I'm trying to run a Python script that reads a CSV file, but it crashes with a 'FileNotFoundError'.*

AI Final answer:

Okay, great! That's a very common problem. Let's break down why your Python script is crashing with a `FileNotFoundError` when trying to read a CSV file.

Here's the prompt:

David Miller

I'm trying to run a Python script that reads a CSV file, but it crashes with a `'FileNotFoundError'`.

Now, to help me understand *exactly* what's going wrong, could you tell me:

1. What is the *exact* Python code you're using to read the CSV file? Please share the code snippet. Even just the beginning of the script is helpful.
2. What is the *entire* contents of your CSV file? (You can paste it here, or provide a link to where it's located).
3. What is the *exact* error message you're getting when the script crashes? (Copy and paste the full error message.)
4. What version of Python are you using? (e.g., Python 3.9, Python 3.10, etc.)

5. What operating system are you using?

How to rate:

"1": strongly prefer Answer A

"5": strongly prefer Answer B

"3": both answers are equally good

Precision and Focus (1-5): 4

Contextual Fit (1-5): 3

Answer Faithfulness (1-5): 3

Correctness (1-5): 3